## Type MS · State or explain safety precautions for a practical

### Answer 1 of 11 · Question 2 part a.ii · 2 marks

2020 NOV · Paper 01 · extract 1 of 2 · find part a.ii in this mark scheme extract

| number | | | |
|---|---|---|---|
| 2 (a) (i) | better transfer of heat to water (1) | | 1 |
| (a) (ii) | any two from<br>• safety goggles/glasses (1)<br>• water may spit (1)<br>• use test tube holder (1)<br>• to avoid burning hands (1) | reason linked to precaution | 2 |
| (b) (i) | • 28 x 12 x 4.2 (1)<br>• 1411/1411.2J (1)<br>• 1400J (1) | error carried forward<br>full marks for correct answer | 3 |
| (ii) | • some energy lost to air/absorbed by tube/needle (1)<br>• some converted to light (1)<br>• food not fully burnt; | | 3 |
| (c) | • use same mass of food (1)<br>• same volume of water (1)<br>• use same thin tube (1)<br>• burn food same distance from tube (1) | | 4 |

Total 13 marks

## Type MS · State or explain safety precautions for a practical

### Answer 1 of 11 · Question 2 part a.ii · 2 marks

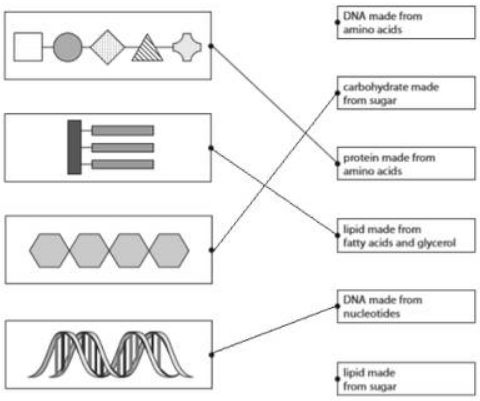
2020 NOV · Paper 01 · extract 2 of 2 · find part a.ii in this mark scheme extract

| Question number | Answer | Notes | Marks |
|---|---|---|---|

Type MS · State or explain safety precautions for a practical

Answer 2 of 11 · Question 3 part d · 2 marks

2019 JUN · Paper 02 · extract 1 of 2 · find part d in this mark scheme extract

| | | | | |
|---|---|---|---|---|
| 3 | (a) |  | Do not allow more than one line from each structure | 3 |
| | (b) | **D** stomach; | | 1 |
| | (c) (i) | 3/three | | 1 |
| | (ii) | not washing equipment/transfer of food containing protein into distilled water; | Allow any valid alternative | 1 |
| | (d) | Hazard: Broken glass - cuts<br>Reducing risk: Clear up breakages using a dustpan and brush/keep equipment away from edge of tables; | Ignore wear gloves | 1 |
| | | Hazard: Biuret reagent - irritant<br>Reducing risk: Wear goggles/wash hands after use/don't use if skin is sensitive; | Ignore wear gloves | 1 |

**Total question 3 = 8 marks**

## Type MS · State or explain safety precautions for a practical

### Answer 2 of 11 · Question 3 part d · 2 marks

2019 JUN · Paper 02 · extract 2 of 2 · find part d in this mark scheme extract

| Question number | Answer | Notes | Marks |
|---|---|---|---|

Type MS · State or explain safety precautions for a practical

Answer 3 of 11 · Question 6 part b.ii · 1 mark

2020 JAN · Paper 01 · find part b.ii in this mark scheme extract

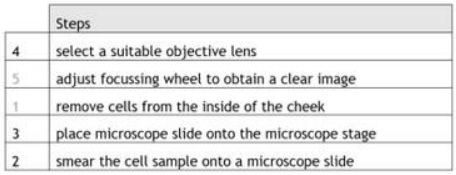
| Number | | | | |
|---|---|---|---|---|
| 6 | (a) | bronchus/bronchi (1) | | 1 |
| | (b) (i) | • inhale and exhale/breathe in an out (1)<br>• through mouthpiece (1)<br>• floating air chamber rises/falls (1)<br>• pen records a trace on (the rotating) graph paper/drum (1) | | 4 |
| | (ii) | Do not share the mouthpiece with another person/disinfect mouthpiece before use by another person/after use (1) | | 1 |
| | (iii) | Any two from:<br>• airways /bronchi/bronchus constricted/obstructed/reduced diameter (1)<br>• less/reduced airflow (1)<br>• per unit time (1) | | 2 |
| | (c) (i) | cannot be transmitted from one person to another/not contagious/not caused by a pathogen/bacteria/virus/microorganism (1) | | 1 |
| | (ii) | 9.5 - 7 = 2.5(1)<br>2.5 ÷ 7 x 100(1)<br>36(%) (1) | Full marks for correct final answer<br>Allow 35.7 for final answer | 3 |
| | (iii) | people taking less exercise/<br>consuming too much fat/sugar/ taking in more energy than is being used/<br>increase in the number of people smoking/drinking more alcohol/obesity (1) | Allow other valid lifestyle factor | 1 |

**Total for Question 6 = 13 marks**

## Type MS · State or explain safety precautions for a practical

### Answer 4 of 11 · Question 3 part a.iv · 2 marks

2020 NOV · Paper 01 (regional R) · find part a.iv in this mark scheme extract

| Number | | | | |
|---|---|---|---|---|
| 3 | (a) (i) |  | One mark for 4 before 5.<br>One mark for 3 before 2 | 2 |
| | (ii) | more detail/organelles can be seen/organelles/named organelle seen more clearly (1) | | 1 |
| | (iii) | reference no stain applied to specimen(1) | | 1 |
| | (iv) | • reference to sterilising (named) equipment (1)<br>• wash hands to avoid cross contamination(1)<br>• incubate/store in sealed petri dish / container | allow valid sterilisation technique | 2 |
| | (b) (i) | 10 × 40 = 400 | Full marks for correct answer without working | 1 |
| | (ii) | reference to 1000 (1)<br>0.018 × 1000 = 18 (µm) | Full marks for correct final answer | 2 |

**Total Question 3 = 9 marks**

## Type MS · State or explain safety precautions for a practical

### Answer 5 of 11 · Question 7 part a.i · 3 marks

2021 JUN · Paper 01 · extract 1 of 2 · find part a.i in this mark scheme extract

| number | | | | |
|---|---|---|---|---|
| 7 | (a) (i) | • safety glasses/goggles;<br>• care using sharp needle;<br>• care to avoid burns;<br>• | | 3 |
| | (ii) | • mass of food;<br>• volume of water;<br>• distance of burning food from tube; | reject amount | 3 |
| | (iii) | any five from<br>• not all food burns to release energy;<br>• not all heat transferred to water;<br>• some lost to atmosphere;<br>• some heat lost to needle/glass tube;<br>• uneven distribution of temperature in water;<br>• because lack of stirring; | | 5 |
| | (b) | any four from<br>• use of oxygen;<br>• allows complete combustion;<br>• enclosed/lid so no heat escapes;<br>• heat transfer coil so all heat is used to increase temp of water/transferred to water;<br>• stirrer to distribute heat; | | 4 |

Total 15 marks

## Type MS · State or explain safety precautions for a practical

### Answer 5 of 11 · Question 7 part a.i · 3 marks

2021 JUN · Paper 01 · extract 2 of 2 · find part a.i in this mark scheme extract

| Question number | Answer | Notes | Marks |
|---|---|---|---|

Type MS · State or explain safety precautions for a practical

Answer 6 of 11 · Question 3 part a.iii · 2 marks

2022 JUN · Paper 01 · extract 1 of 2 · find part a.iii in this mark scheme extract

| | | | |
|---|---|---|---|
| 3 (a) (i) | Any five from the following:<br>• place water in the test tube;<br>• take the (starting) temperature of the water;<br>• attach the test tube to a clamp stand;<br>• measure the mass of a piece of food/weigh food;<br>• place food on the mounted needle (under the test tube);<br>• burn the food; (using a Bunsen burner)<br>• take the (final) temperature of the water; | | 5 |
| (ii) | Any two from the following:<br>• volume/mass of water;<br>• distance of the food from the test tube;<br>• mass of food; | ignore amount<br><br>ignore amount/size | 2 |
| (iii) | Any two from the following:<br>• hold the burning food at arm's length;<br>• reference to sharpness of needle;<br>• goggles;<br>• reference to food allergies/don't taste food; | any ref to being burnt/hot water | 2 |
| (b) (i) | Any two from the following:<br>• males need more energy than females;<br>• the amount of energy required decreases with age;<br>• the amount of energy needed by males decreases more (overall) than females; | | 2 |
| (ii) | **D** - Males use more muscle mass than females; | | 1 |
| (iii) | • (2629 – 2103) = 526;<br>• 526 ÷ 2103;<br>• (x 100) = 25 (%); | if divide by 2629 and give 20% then two marks<br><br>Total 15 marks | 3 |

## Type MS · State or explain safety precautions for a practical

### Answer 6 of 11 · Question 3 part a.iii · 2 marks

2022 JUN · Paper 01 · extract 2 of 2 · find part a.iii in this mark scheme extract

| Question number | Answer | Notes | Marks |
|---|---|---|---|
| | | | |

## Type MS · State or explain safety precautions for a practical

### Answer 7 of 11 · Question 1 part d · 1 mark

2022 JUN · Paper 02 (regional R) · extract 1 of 2 · find part d in this mark scheme extract

| | | | |
|---|---|---|---|
| 1 (a) | • apparatus in correct order with T piece connected to both flasks;<br>• long tubes in indicator;<br>• short tubes above indicator; | | 3 |
| (b) | limewater; | Allow sodium hydrogen carbonate | 1 |
| (c) | any four from:<br>• breathe through mouth piece;<br>• in and out;<br>• 10 times/max of one minute;<br>• compare contents of flasks;<br>• exhaled flask should go cloudy if more carbon dioxide exhaled; | | 4 |
| (d) | sterilise mouth piece/breath gently; | | 1 |
| (e) | same volume of indicator in each flask; | | 1 |

Total 10 marks

## Type MS · State or explain safety precautions for a practical

### Answer 7 of 11 · Question 1 part d · 1 mark

2022 JUN · Paper 02 (regional R) · extract 2 of 2 · find part d in this mark scheme extract

| Question number | Answer | Notes | Marks |
|---|---|---|---|

## Type MS · State or explain safety precautions for a practical

### Answer 8 of 11 · Question 1 part c.i · 1 mark

2024 JUN · Paper 01 · find part c.i in this mark scheme extract

| | | | | |
|---|---|---|---|---|
| 1 (a) | (i) | sensory (neurone/cell); | | (1) |
| | (ii) | axon; | | (1) |
| (b) | (i) | Part R sweat gland;<br>Part S blood vessels/capillaries/arterioles; | | (2) |
| | (ii) | Any two from:<br>• produces sweat/perspiration;<br>• sweat evaporates (from skins surface);<br>• using heat energy (from the blood); | | (2) |
| | (iii) | • provides an insulating layer insulation;<br>• traps heat (energy)/to prevent heat loss; | reject keeps body warm | (2) |
| (c) | (i) | • water is too hot/burn hands; | | (1) |
| | (ii) | • use a thermometer;<br>• to test the temperature of the water/to make sure the water is not too hot; | | (2) |
| | (iii) | • (temperature) receptor; | | (1) |

**Total for Question 1 = 12 Marks**

## Type MS · State or explain safety precautions for a practical

### Answer 9 of 11 · Question 1 part c.ii · 2 marks

2024 JUN · Paper 01 · find part c.ii in this mark scheme extract

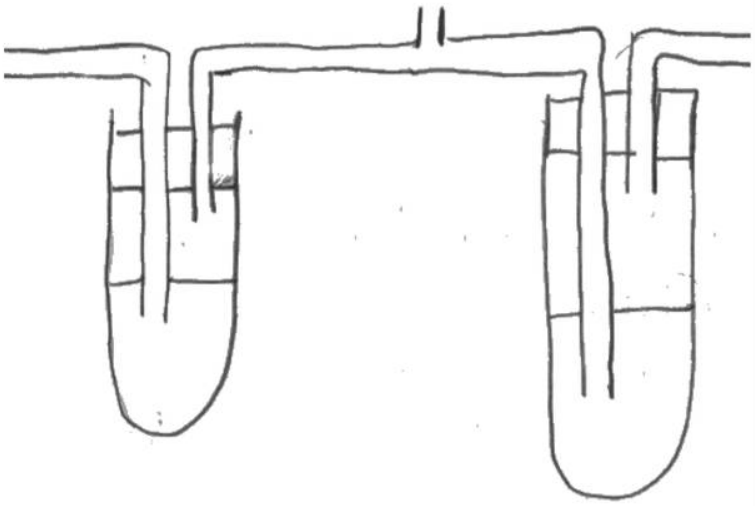
| | | | | |
|---|---|---|---|---|
| 1 (a) | (i) | sensory (neurone/cell); | | (1) |
| | (ii) | axon; | | (1) |
| (b) | (i) | Part R sweat gland;<br>Part S blood vessels/capillaries/arterioles; | | (2) |
| | (ii) | Any two from:<br>• produces sweat/perspiration;<br>• sweat evaporates (from skins surface);<br>• using heat energy (from the blood); | | (2) |
| | (iii) | • provides an insulating layer insulation;<br>• traps heat (energy)/to prevent heat loss; | reject keeps body warm | (2) |
| (c) | (i) | • water is too hot/burn hands; | | (1) |
| | (ii) | • use a thermometer;<br>• to test the temperature of the water/to make sure the water is not too hot; | | (2) |
| | (iii) | • (temperature) receptor; | | (1) |

**Total for Question 1 = 12 Marks**

Type MS · State or explain safety precautions for a practical

Answer 10 of 11 · Question 4 part a.ii · 1 mark

2024 JUN · Paper 01 · find part a.ii in this mark scheme extract

| | | | | | |
|---|---|---|---|---|---|
| 4 | (a) | (i) | • apparatus correctly set up;<br>• long tube in indicator;<br>• short tube above indicator;<br>  | | (3) |
| | | (ii) | Inhale and exhale gently/sterilise/clean mouthpiece/don't inhale liquid/care in assembling apparatus to avoid breaking glass; | | (1) |
| | (b) | (i) | exercise/rest for the same length of time/at the same pace for each repeat/same student; | | (1) |
| | | (ii) | any four from<br>• breathing rate increases (with vigorous exercise);<br>• more oxygen inhaled/supplied;<br>• for aerobic respiration;<br>• more energy released/more ATP;<br>• breakdown of lactic acid;<br>• produced by anaerobic respiration; | ignore ref $CO_2$ | (4) |

**Total for Question 4 = 9 marks**

## Type MS · State or explain safety precautions for a practical

### Answer 11 of 11 · Question 8 part b.ii · 1 mark

2025 JUN · Paper 02 (regional R) · find part b.ii in this mark scheme extract

| | | | | |
|---|---|---|---|---|
| 8 (a) | (i) | cilia/skin/mucus | allow other suitable answers | 1 |
| | (ii) | kills/inhibits growth of pathogens/damage cell membranes; | | 1 |
| | (iii) | • (phagocytes) engulf pathogens/microorganisms / bacteria;<br>• digest/breakdown pathogens/microorganisms / bacteria; | | 2 |
| | (iv) | antigen/immunogen; | | 1 |
| | (v) | • antibodies have a particular shape /are complementary (to antigen);<br>• only recognise/attach/bind to, one particular/specific pathogen / antigen/receptor; | | 2 |
| | (vi) | • natural passive immunity the body receives antibodies naturally;<br>• artificial passive immunity are when antibodies are injected into the body; | accept through breast milk/placenta<br>accept e.g. to give immediate protection against a disease; | 2 |
| (b) | (i) | 360 ÷ 30 = 12;<br>108 ÷ 12;<br>9; | | 3 |
| | (ii) | reduce the risk of disease transmission/infection; | | 1 |
| | (iii) | any two from<br>• improper/overuse of antibiotics/over prescribed;<br>• bacterial DNA mutates/bacteria adapt;<br>• some survive and reproduce | | 2 |
| | | | **Total for Question 8 = 15 marks** | |